

GrapeEngine Level Editor Guide

This guide explains the usage of the **editor panels**:

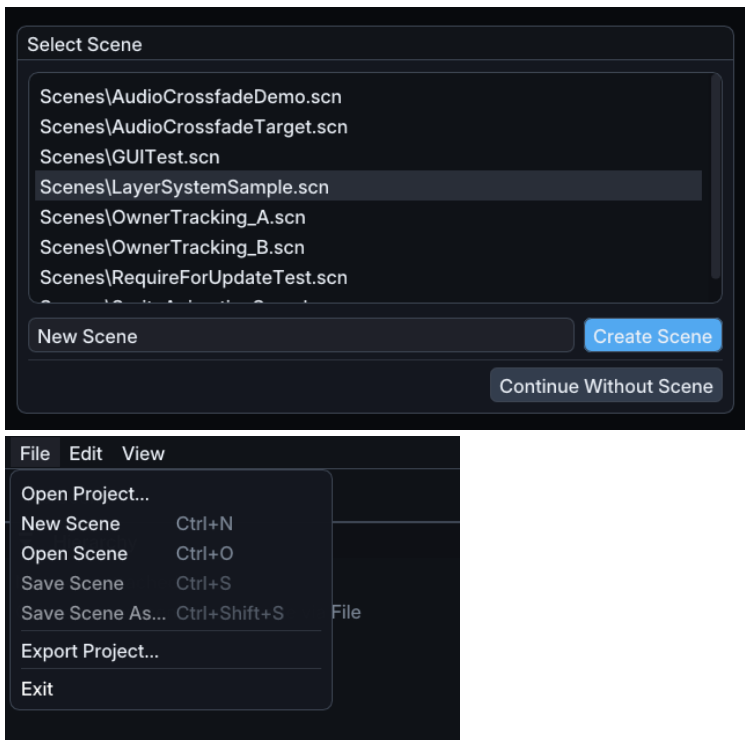
- Hierarchy
- Property/Prefab Editor
- Asset Browser
- Scene and Game Viewports
- Playback Controls
- Tile Palette
- Layers
- Systems
- Console
- Performance
- Game Configuration

It also covers **workflows** such as project startup, scene/entity editing, component and prefab editing, asset import and drag-and-drop, tilemap painting and collision editing, playtesting and save/export shortcuts.

Project Startup

1. In the startup screen, click **Add from Disk** and select the **TestSuite** folder
2. Open the project by either:
 - Double-clicking the added project, or
 - Selecting it once and clicking **Open Project**
3. After opening, choose how to enter the editor:
 - Double-click an existing scene
 - Change the name in the input text field (optional) and click **Create Scene**, then double-click the added scene
 - Click **Continue Without Scene**
4. If you choose **Continue Without Scene**, use **File > New Scene** after entering the editor
5. Otherwise, the editor is ready for use immediately

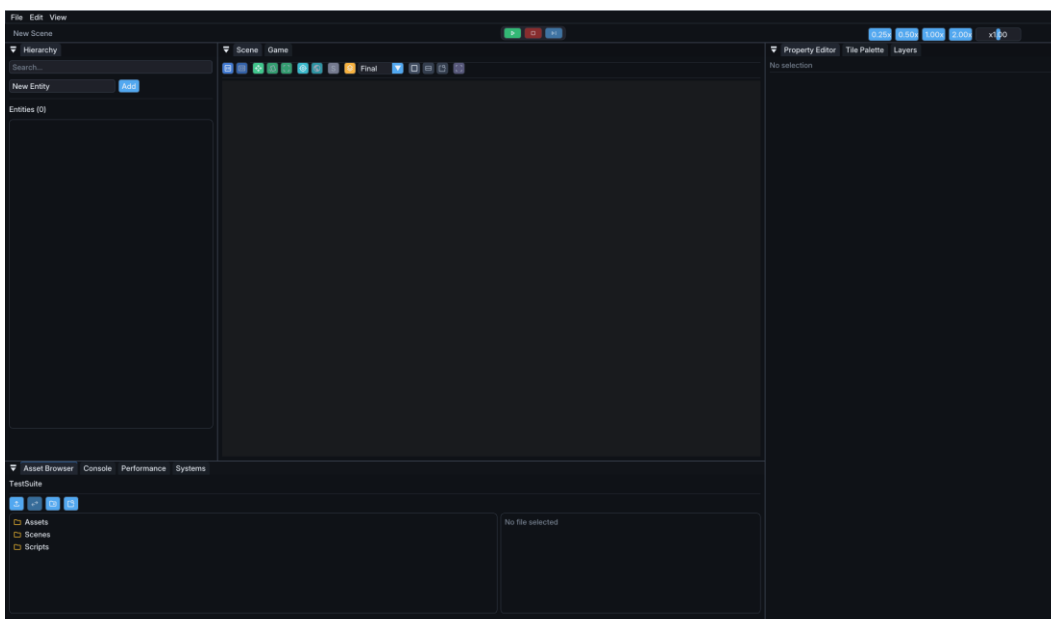
Scene selection:



Editor Layout & Docking

The editor uses ImGui docking to organize panels:

- Top toolbar: Game Controls (Play/Pause/Stop, Step)
- Left: Hierarchy
- Center: Scene/Game Viewports
- Right: Property/Prefab Editor, Tile Palette, Layers
- Bottom: Asset Browser, Console, Performance, Systems



Hierarchy

The panel displays scene entities in a parent/child tree and keeps selection synced with the Scene viewport.

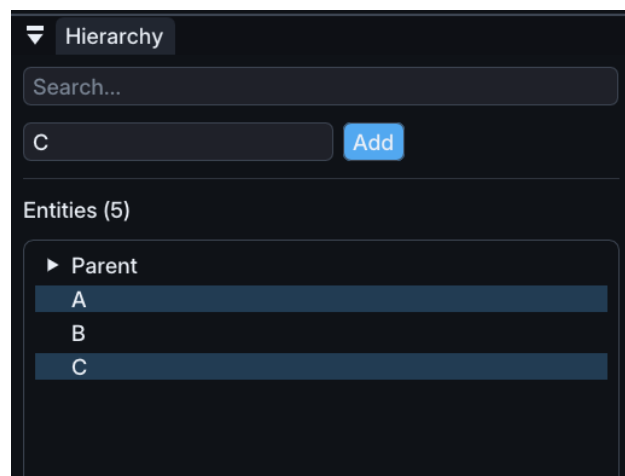
Create a New Entity

1. Open Hierarchy panel
2. Enter a name in the input field (defaults to "New Entity")
3. Click Add to create a root-level entity
4. Drag the entity to reparent under another node

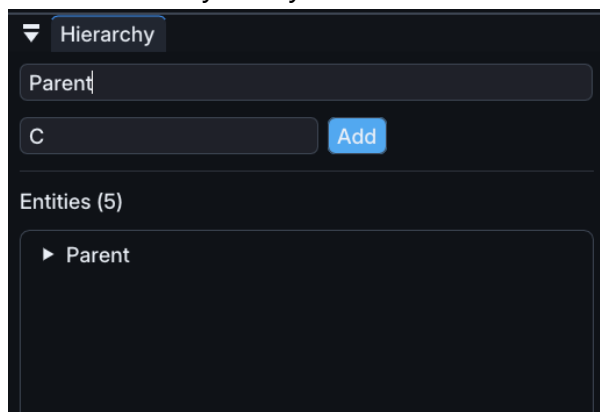


It supports:

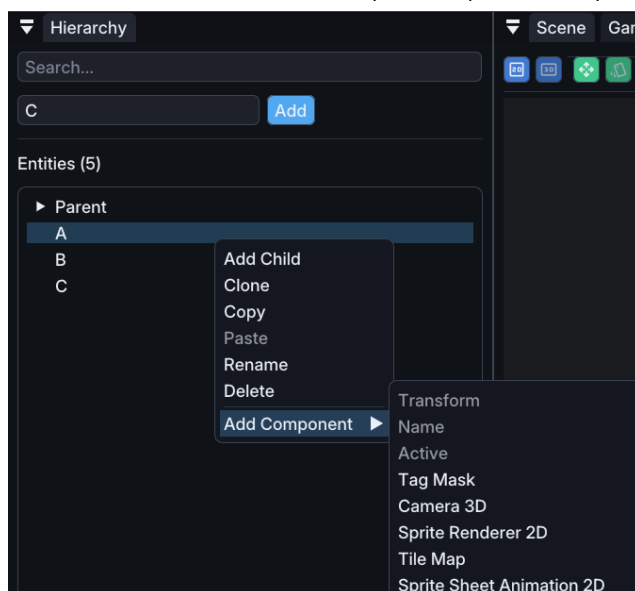
- Single selection
- Multi-selection (Ctrl+Click/Shift+Click)



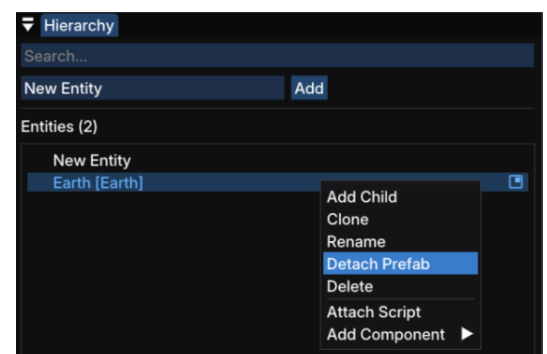
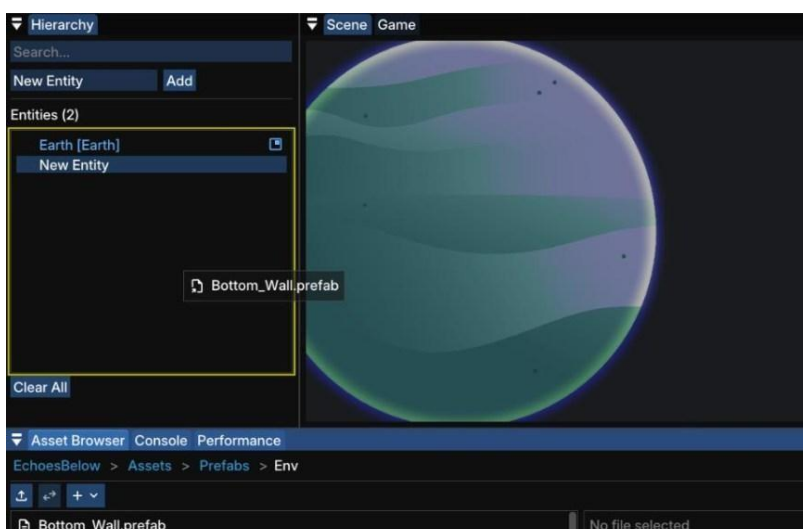
- Search/filter by entity name



- Create root entities (including GUI presets)
- Context actions: Add Child, Clone, Rename, Delete, Detach Prefab and Add Component



- Drag-and-drop reparenting and sibling reordering
- Prefab instantiation by dragging .prefab assets onto an entity (as child) or empty tree space (as root)

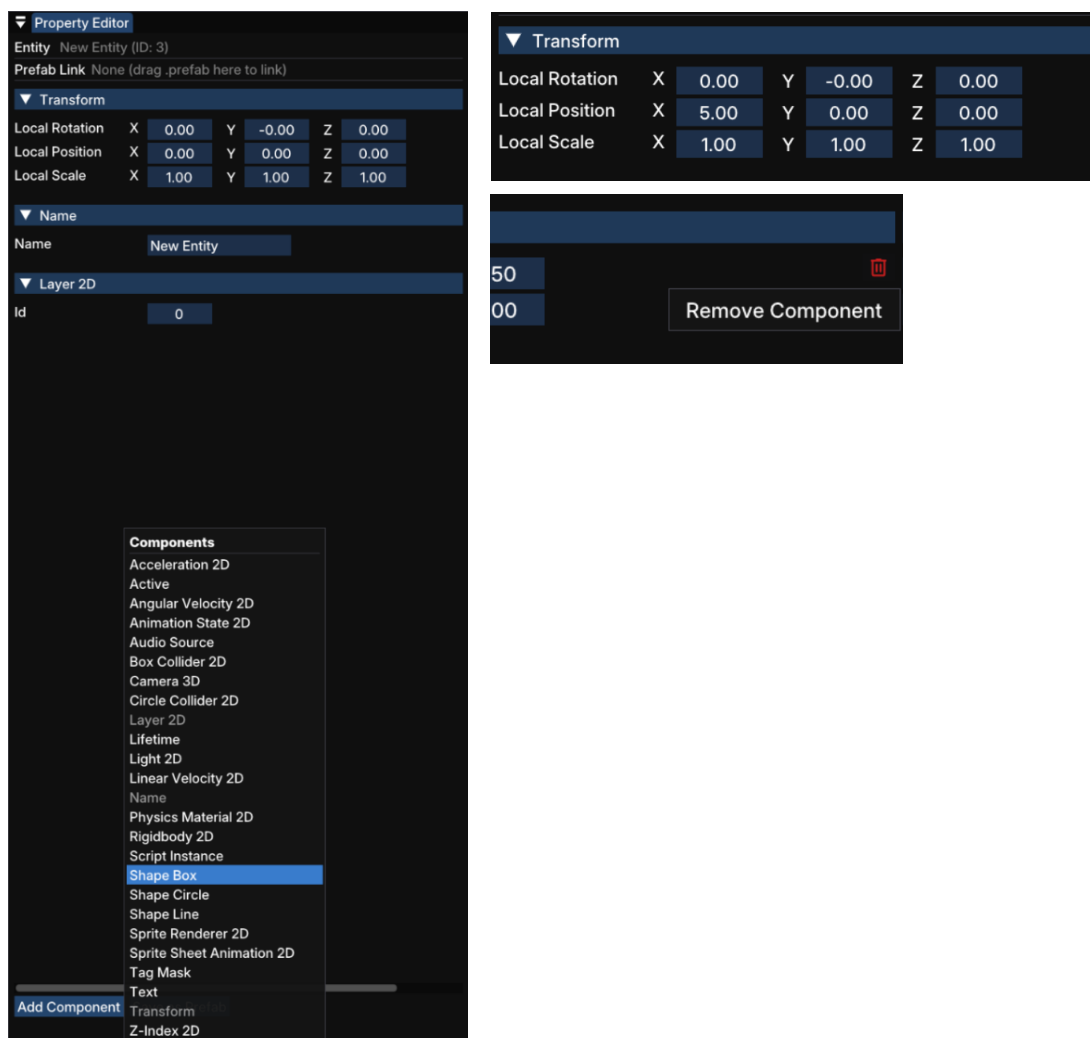


- Clipboard shortcuts: copy (Ctrl+C)/cut (Ctrl+X)/paste entities (Ctrl+V)
- Rename shortcut (F2) and delete shortcut (Del)
- Undo/redo integration for hierarchy operations (create/delete/reorder, etc.)

Property/Prefab Editor

This panel lets you select an entity (in **Hierarchy**) or a prefab (in **Hierarchy** or **Asset Browser**) to inspect and edit components.

Use **Add Component** (with search) to add new components. Most edits apply immediately in-scene and are written back after the UI pass.



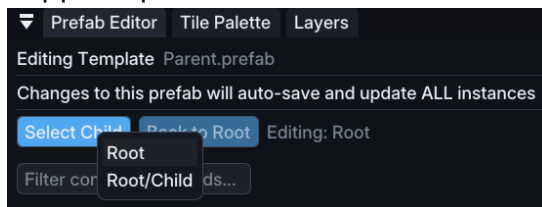
Dragging a root prefab from **Asset Browser** to **Hierarchy** creates a prefab instance. Just like any other entity, its components can be modified/added/removed.

Supported features:

- Multi-selection editing
- Undo/redo tracks property edits and component add/remove/reset operations (not just transform)
- Component list supports filtering (Ctrl+F), and Add Component popup can be opened via shortcut (Ctrl+Shift+A)
- Component deletion

Prefab Editor Mode

1. Open a .prefab (root prefab) from Asset Browser
2. Edit prefab node components, including add/remove/reset
3. Save prefab changes and Apply to All Instances in the current world to apply changes to all prefab instances
4. Supports prefab node selection when editing prefab hierarchies



Asset Browser

This panel lets you browse assets, import new files and drag-drop into components.

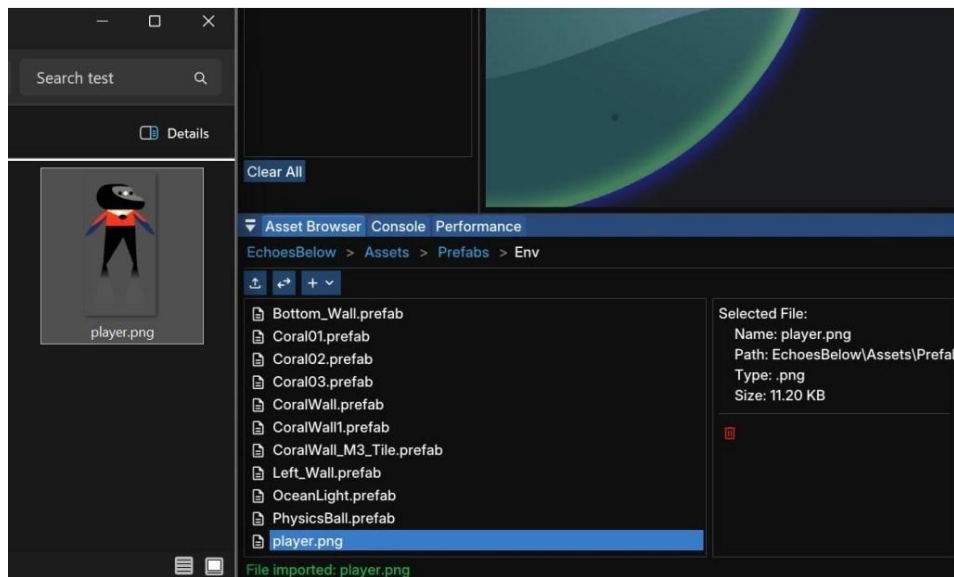
Internal drag-drop:

- Drag textures/audio and other supported assets onto matching component fields
- Drag assets between folders in Asset Browser (Ctrl to copy, otherwise move)



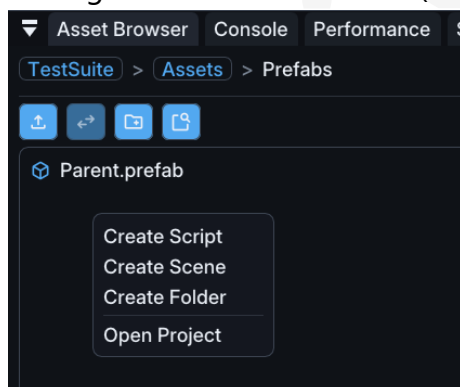
External drop:

- Drag files from the Windows Explorer into the editor window to import into the current project folder



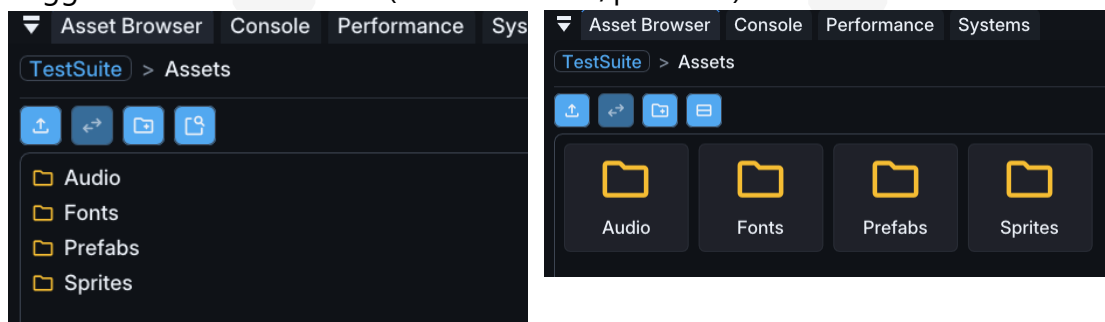
Create assets:

- Use right-click or the + menu (Create Script, Create Scene, Create Folder)



Navigation and views:

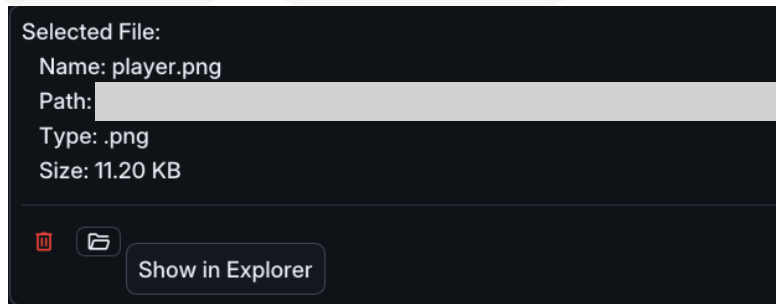
- Breadcrumb navigation at the top
- Toggle between Grid view (tile thumbnails/previews) and List view



Context menu/shortcuts:

- Rename (F2), Copy (Ctrl+C), Cut (Ctrl+X), Paste (Ctrl+V), Delete (Del)
- Prefabs: Load Prefab into scene, Edit Prefab in Prefab Editor

- Open Project and Show in Explorer actions are supported

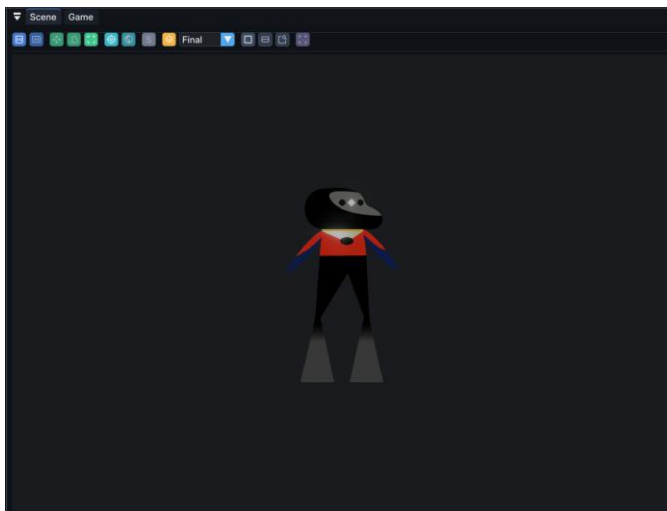


Scene/Game Viewport

The Scene viewport shows the editor camera view for editing, while the Game viewport shows output from the active scene Camera3D.

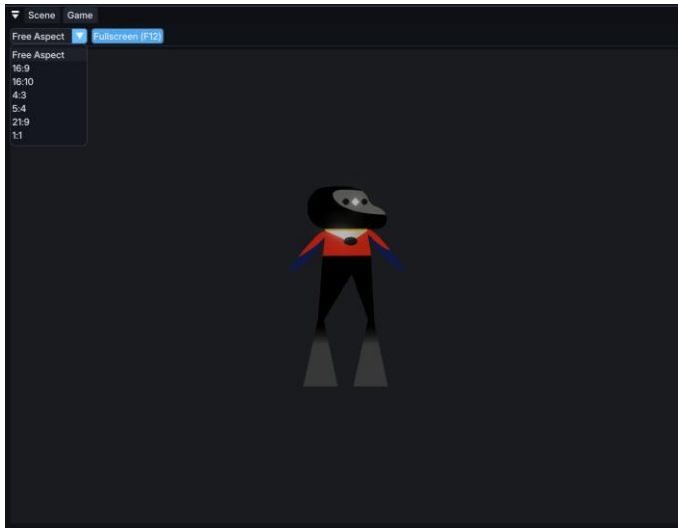
Scene Viewport

- Supports selection/picking, plus gizmo editing: Move (T)/Rotate (E)/Scale (R), local/world mode, snap settings and focus selected (F)
- Selection in Scene viewport syncs with Hierarchy/Property Editor



Game Viewport

- Supports aspect-ratio presets, free aspect, and immersive fullscreen (F12, Esc to exit)
- Create a new entity and attach the Camera3D component; tick the Active checkbox
- Change the Z-position to a value like 10 to see other entities in the Game viewport



Playback Controls

The panel provides the following functionalities:

- **Play:** Saves a snapshot and starts simulation
- **Pause:** Freezes time (timeScale = 0)
- **Stop:** Restores snapshot; returns to edit state
- **Step:** Advances one frame while paused

Hotkeys: Ctrl + P (Play/Stop), Ctrl+Shift+P (Pause/Resume), Alt+P (Step)

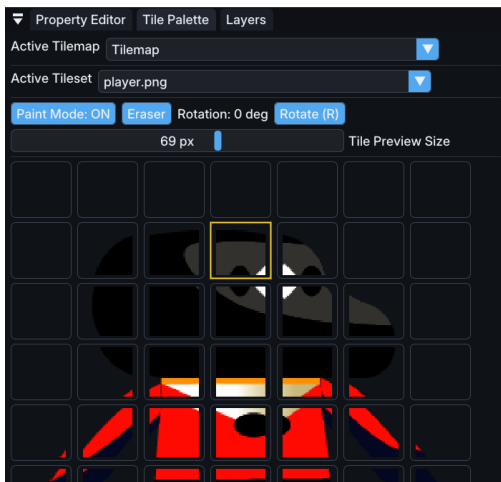


Tile Palette

This panel is an editor tool for painting tiles onto a tilemap in the game engine. What it does is it lets the user select tiles from a palette and paint them into the level by clicking in the viewport. Collisions can also be added to painted tiles in the same panel.

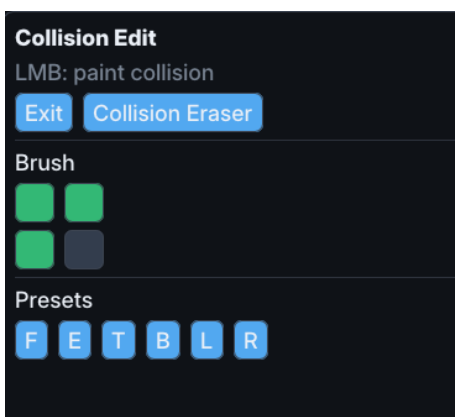
Tile Painting/Erasing

- Drag a .png or .tilemap asset to **Tile Palette**
- Click a tile in the panel and left-click in the viewport to paint that tile (**Paint Mode** = on)
- Toggle **Eraser** = on to delete tiles
- Change the rotation (0-3) before painting if the tile needs to be rotated
- Play with the tile preview size slider to resize the tiles to your liking



Collision Painting/Erasing

- One way to access this is by clicking on the **Tilemap** entity in **Hierarchy**; under the **Tile Map** component, click **Edit Collision**
- Another way is to click on the **Collision Edit** child window in **Tile Palette**
- Left-click in the viewport to paint collision
- Toggle **Collision Eraser** = on to clear collision
- Use 2x2 corner brush controls and presets (Full/Empty/Top/Bottom/Left/Right)
- Works only on occupied, i.e. painted, tiles
- **Must click Exit to resume painting tiles**

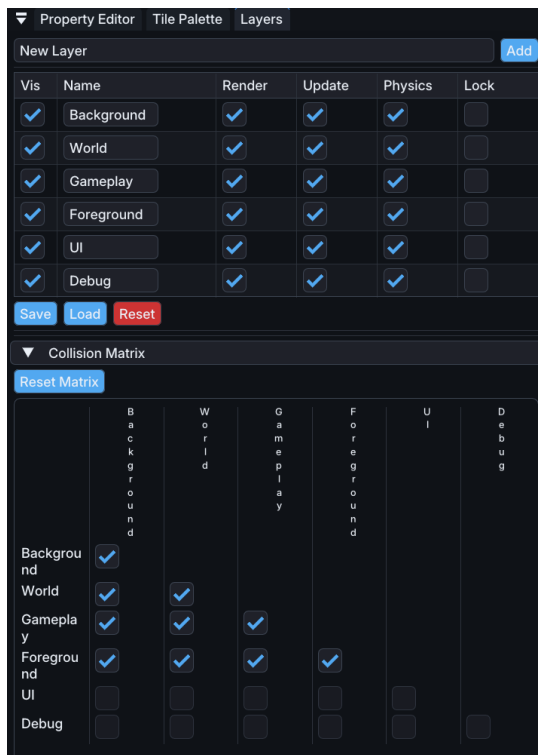


Layers

This panel manages scene layers and collision relationships:

- Create, rename, reorder, and remove layers
- Toggle per-layer states: visibility, render, update, physics, and lock
- Assign selected entities to layers
- Edit the layer collision matrix (including reset/import options)

- Supports save/load/reset actions for layer configuration



Systems

This panel shows all registered ECS systems and their runtime metadata:

- Lists systems by execution group (PreUpdate/Update/PostUpdate, Physics, Render, etc.)
- Displays system type (Native vs C#), enabled state, and edit-mode execution status
- Lets you toggle individual systems on/off at runtime
- Includes summary stats (counts by type/enabled/edit-mode)

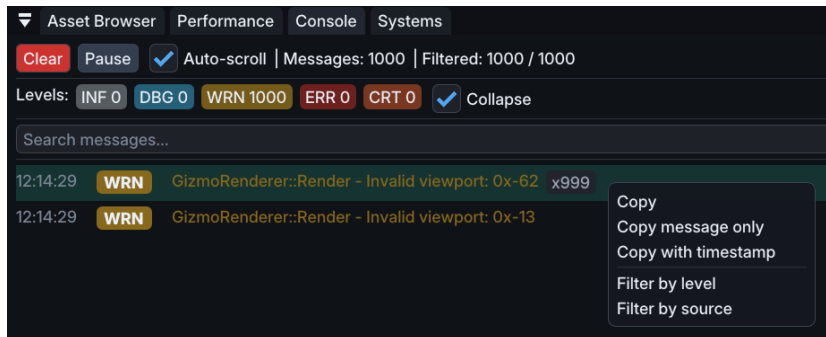
Registered ECS Systems		
System Name	Group	Type
Update (5)		
Scripts.RequireForUpdateTestSystem	Update	[C#] Scripts.RequireForUpdateTestSystem
BoidSystem	Update	[Native] BoidSystem
ParticleSystem	Update	[Native] ParticleSystem
AnimationPreview	Update	[Native] AnimationPreview
Animation	Update	[Native] Animation

Console

This panel shows engine/editor logs in real time for debugging and workflow feedback:

- Displays messages by log level with filtering/search (INF, DBG, WRN, ERR, CRT)

- Supports selecting/copying logs (including copy-all/copy-filtered actions)
- Context actions include copy variants and clear
- Works with keyboard shortcuts when focused (for example Ctrl+A select all)



Performance

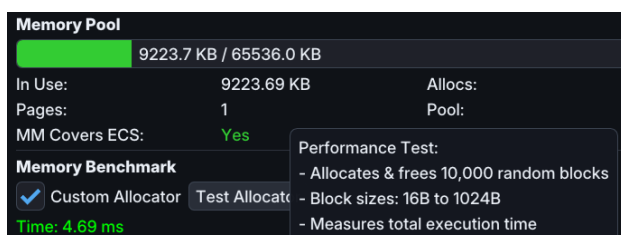
This panel displays runtime/editor performance stats to help profile scene behavior:

- Shows frame timing and FPS metrics
- Updates live while scene/editor is running
- Resets appropriately when switching scenes



Memory Benchmark

- Custom Allocator toggle
- Run Test button (10,000 alloc/free ops) to compare/test between the default allocator and the custom one the engine uses
- Live memory pool stats (allocated/freed/usage/page/free blocks)

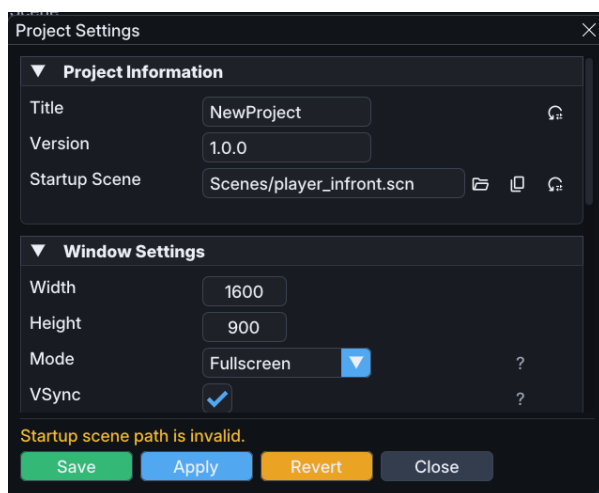


Game Configuration

Lets you configure project-level runtime settings used by the game build:

- Open via **Edit > Project Settings...**
- Sections include:
 - Project Information (Title, Version, Startup Scene)
 - Window Settings (Width, Height, Mode: Windowed/Fullscreen/Borderless, VSync)
 - Physics Settings (Gravity, TimeStep)
 - Audio Settings (Master, Music, SFX, Mute When Unfocused)
 - Performance Settings (Max FPS when VSync is off)
- Per-field and per-section reset controls are available
- **Save, Revert,** and unsaved-changes confirmation are supported

Take note: Apply the startup scene before exporting

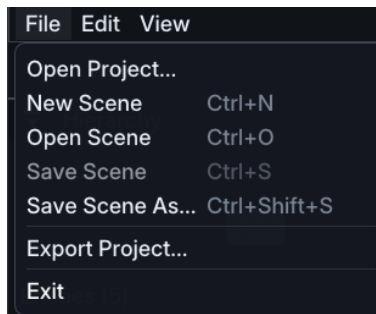


Save/Export

The editor supports scene saving and full project export workflows.

Scene Save Flow

- **File > Save Scene** (Ctrl+S)
- **Save Scene As...** (Ctrl+Shift+S)
- Save is edit-mode only and disabled during play
- Preserves hierarchy/entity order and scene-linked data (including tilemap asset saves)



Project Export Flow

- File > Export Project...
- **Take note: Apply the startup scene before exporting**
- Prompts for destination folder
- Runs build/export pipeline (configure, build, script compile, copy settings/output)
- Shows progress and final summary (success/failure per step + details)

